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NITTE (Deemed to be University)

(Established under Sec.(3) of UGC Act, 1956)

First Semester M.Tech (CBCS) Degree Examinations, December 2025

25MCT102 – FUNDAMENTALS OF MECHATRONICS SYSTEMS

(For MCT)

Duration: 3 Hours]

[Max. Marks: 100

Note: Answer **Five full** questions choosing **One full** question from **each Module**.

Module – I

	Marks	BT*	CO*	PO*
1. a) Define mechatronics system. Explain briefly the various components of a mechatronic system with the help of a block diagram	6	2	1	1
b) Apply the principles of sequential control to develop a control sequence for controlling various operations of domestic automatic washing machine and represent it schematically.	10	3	1	1
c) Explain the working principle of a digital thermometer	4	2	1	1
2. a) What are photo diodes? Briefly explain the working principle of photodiodes	8	2	1	1
b) Write the difference between primary and secondary transducers with examples	4	1	1	1
c) Sketch and explain the working of a hall effect sensor	8	2	1	1

Module – II

3. a) Define a signal. Describe the following basic elementary signals using mathematical expressions and plots (i) Exponential signal (ii) Unit parabolic signal (iii) Signum function	7	1,3	2	1
b) Briefly explain the various effects of time delay and the various ways of handling time delays	8	2	2	1
c) Apply first order Taylor series to linearize a scalar nonlinear state equation represented by $\dot{x} = (-x^3 + u)$ about equilibrium point (0,0).	5	3	2	2
4. a) Explain the procedure and processes involved in analog to digital conversion process with the help of a block diagram	8	2	2	1
b) Prove that the differential amplifier voltage output is the measure of difference between two input voltages	6	3	2	2
c) Briefly explain the virtual instrumentation system using block diagram	6	2	2	1

Module – III

5. a) Draw the general internal architecture of a microprocessor and briefly explain the functions of various constituent parts of a microprocessor	10	2	3	1
b) Explain briefly the working principle of various micro sensors	10	2	3	1
6. a) Briefly explain thermal micro actuator and MEMS micro actuator along with their applications	10	2	3	1
b) Discuss in detail the interaction of materials and micro physics in reference with micro mechatronic systems	10	3	3	1

Module – IV

7. a) Explain the working principle of a hydraulic system with a neat block diagram	10	2	4	1
b) Explain with sketch the construction and working of a hybrid stepper motor	10	2	4	1

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8. a) With a neat diagram explain the torque speed characteristic of a stepper motor.
b) Explain real time control system and differentiate between hard real time and soft real time control system

10	2	4	1
10	2	4	1

Module – V

9. a) Describe the basic components of a closed loop control system with a neat block diagram
b) Define stability of a control system. Analyze the stability of a control system represented by characteristic equation $s^6 + s^5 + 5s^4 + 3s^3 + 2s^2 - 4s - 8 = 0$ using Routh array stability criterion.
10. a) Differentiate between P, PI and PID controllers
b) Consider a unity feedback system with open loop transfer function $G(s) = \frac{5}{s(s+0.5)(s+1)}$. Design a PD controller so that the phase margin of the system is 30° at a frequency of 1.2 rad/sec.

10	2	5	1
10	4	5	2
8	1	5	1
12	4	5	2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
